

AZURE BLOB STORAGE AND DATA FACTORY IMPLEMENTATION

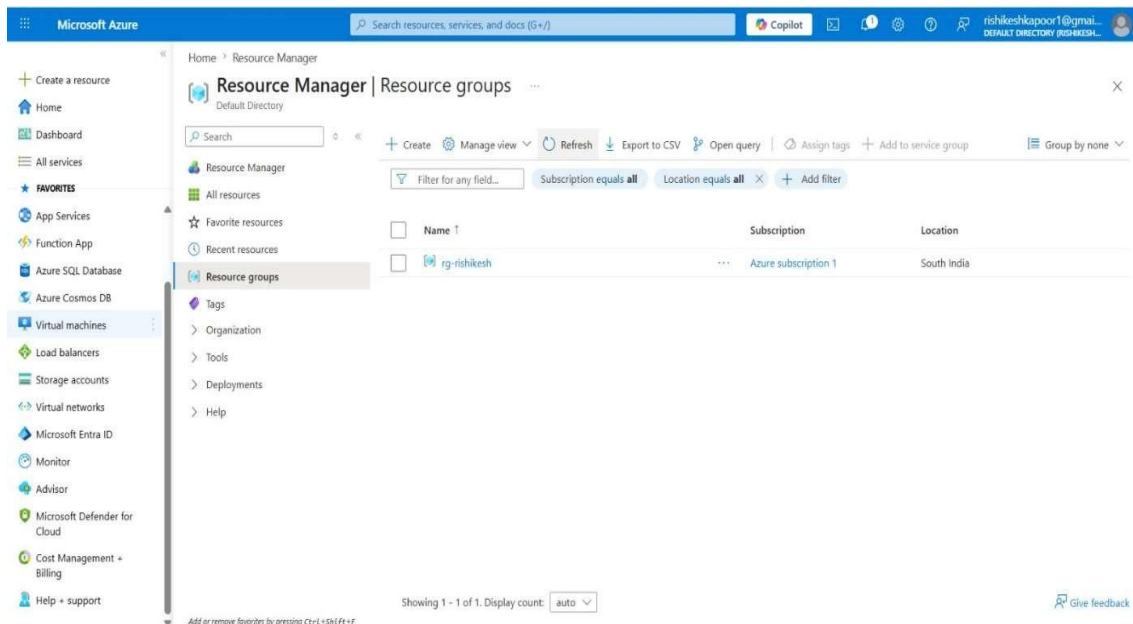
Overview

This assignment demonstrates the use of Azure Blob Storage and Azure Data Factory to transfer data from one storage container to another. It also explains the implementation of Identity and Access Management (IAM) roles and the use of the Get Metadata activity to retrieve file information before initiating the data transfer process.

Step 1 – Setting Up the Resource Group

Configuration Details

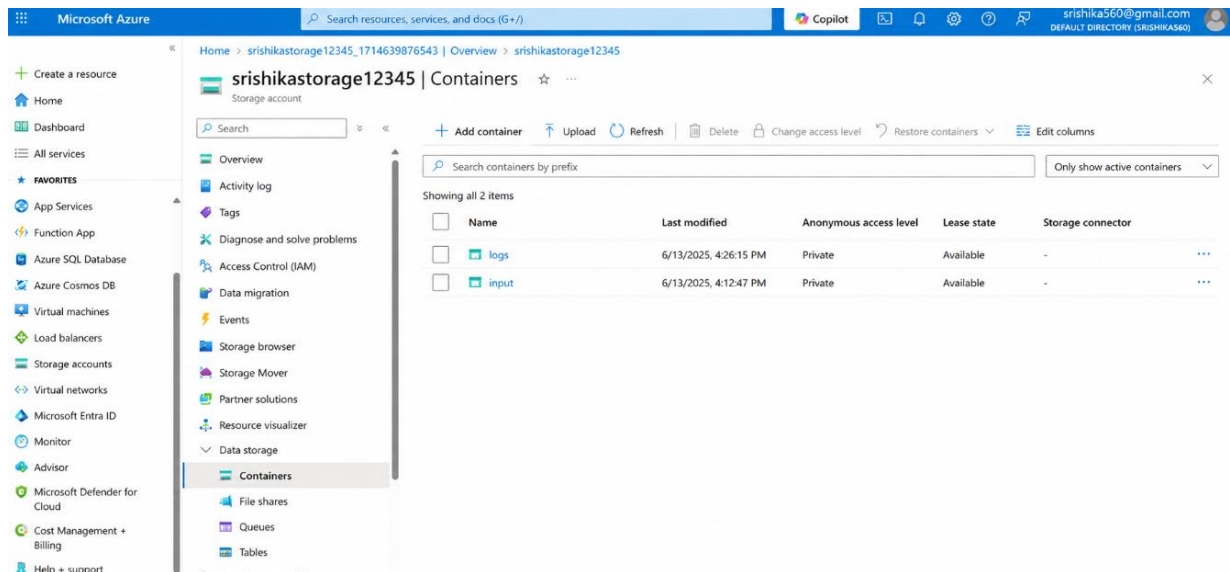
- Resource Group Name: rg-rishika
- Region: South India



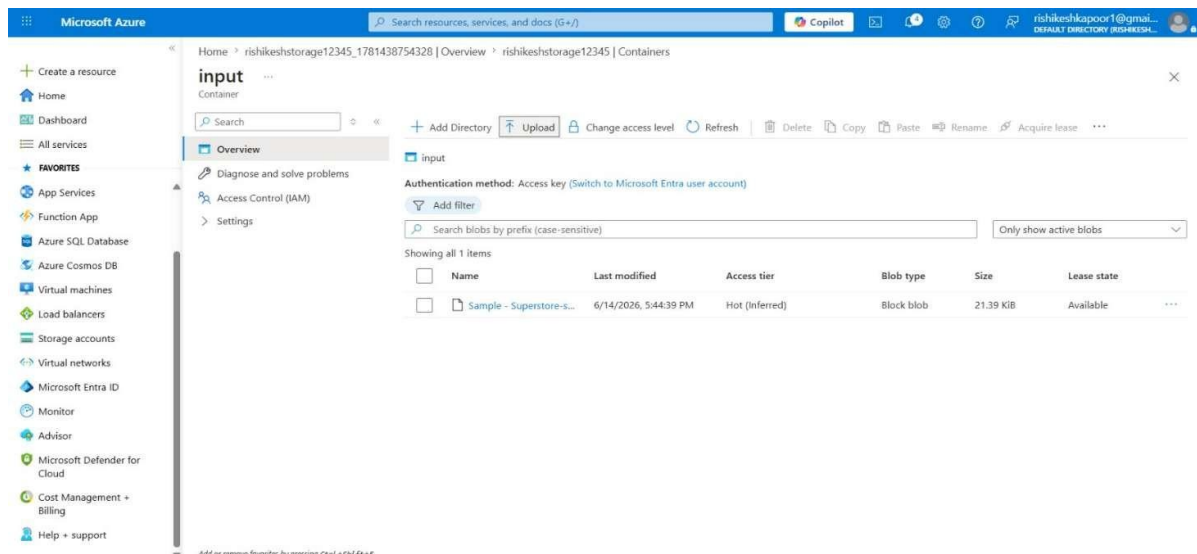
Step 2 – Configuring the Storage Account

Configuration

- Storage Account Name: srishikastorage12345
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- Region: South India



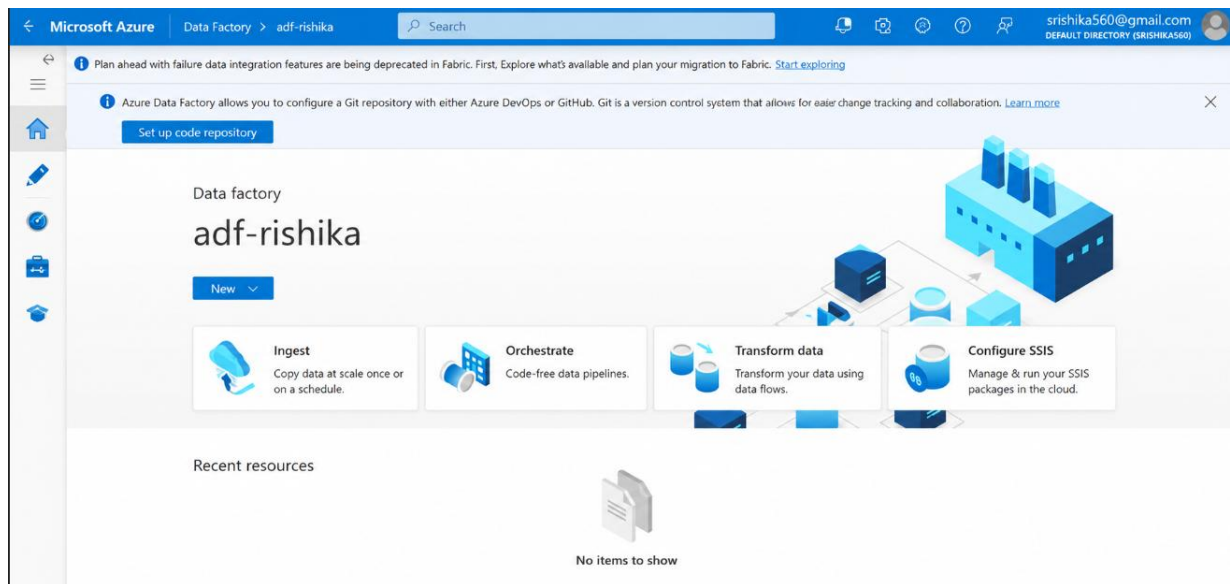
After creating the storage account, the source CSV file was uploaded to the input container.



TASK 3 – Azure Data Factory Basics

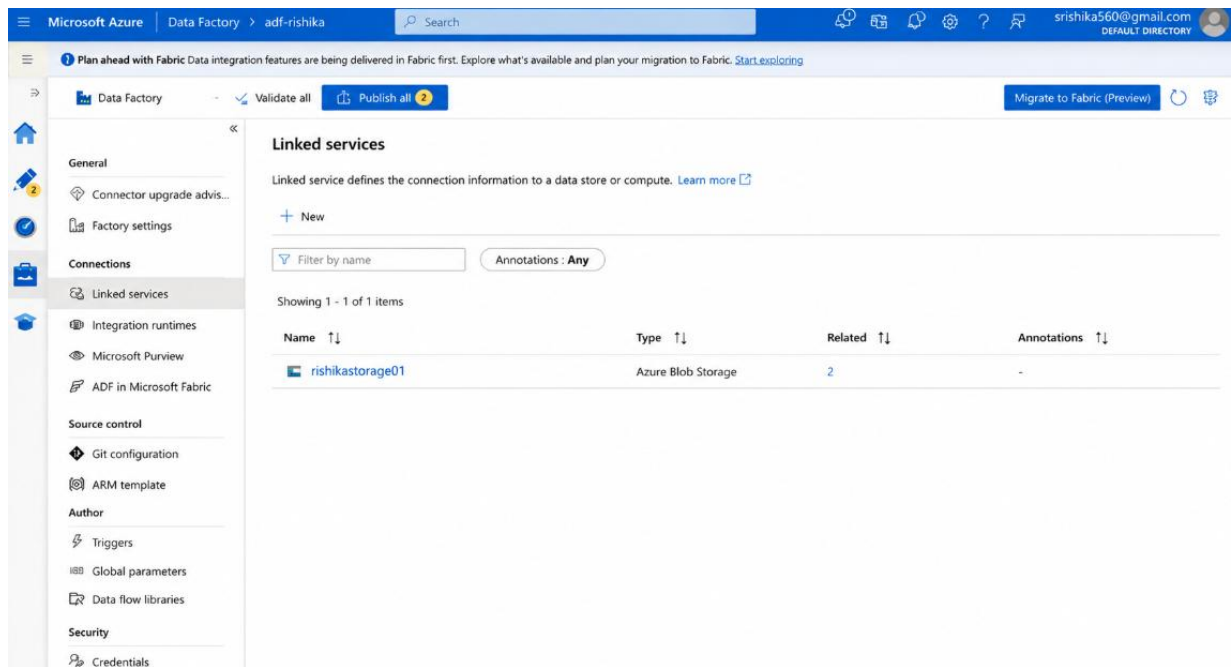
Creating the Azure Data Factory Instance

Azure Data Factory (ADF) is a cloud-based data integration service provided by Microsoft Azure. It enables users to build, schedule, and manage data pipelines that can move and transform data across various storage services and platforms.



❑ Establishing a Linked Service

A Linked Service acts as the connection between Azure Data Factory and external data sources. In this assignment, a linked service named **rishikastorage01** was configured to connect Azure Data Factory with the Azure Blob Storage account. This connection allows the pipeline to securely access the files stored in the storage account



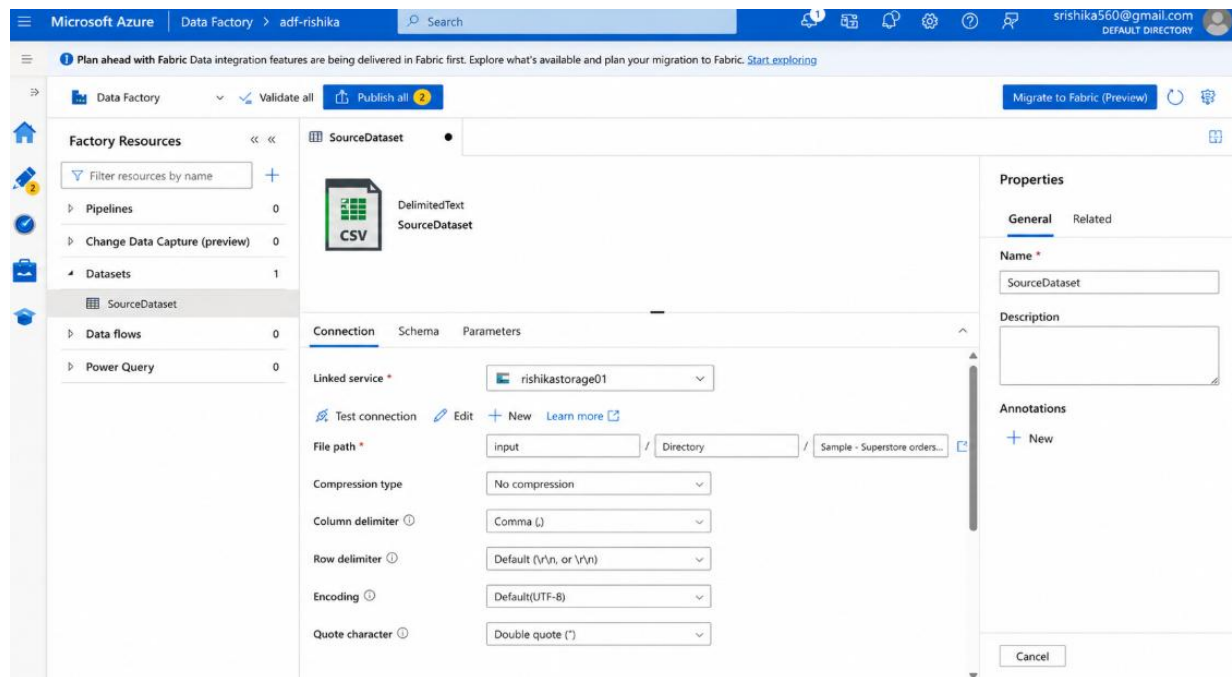
□ Defining Source and Destination Datasets

A Dataset specifies the structure and location of the data used within a pipeline.

Input Dataset

- **Name:** SourceDataset
- **Container:** input
- **File:** Sample-Superstore.csv

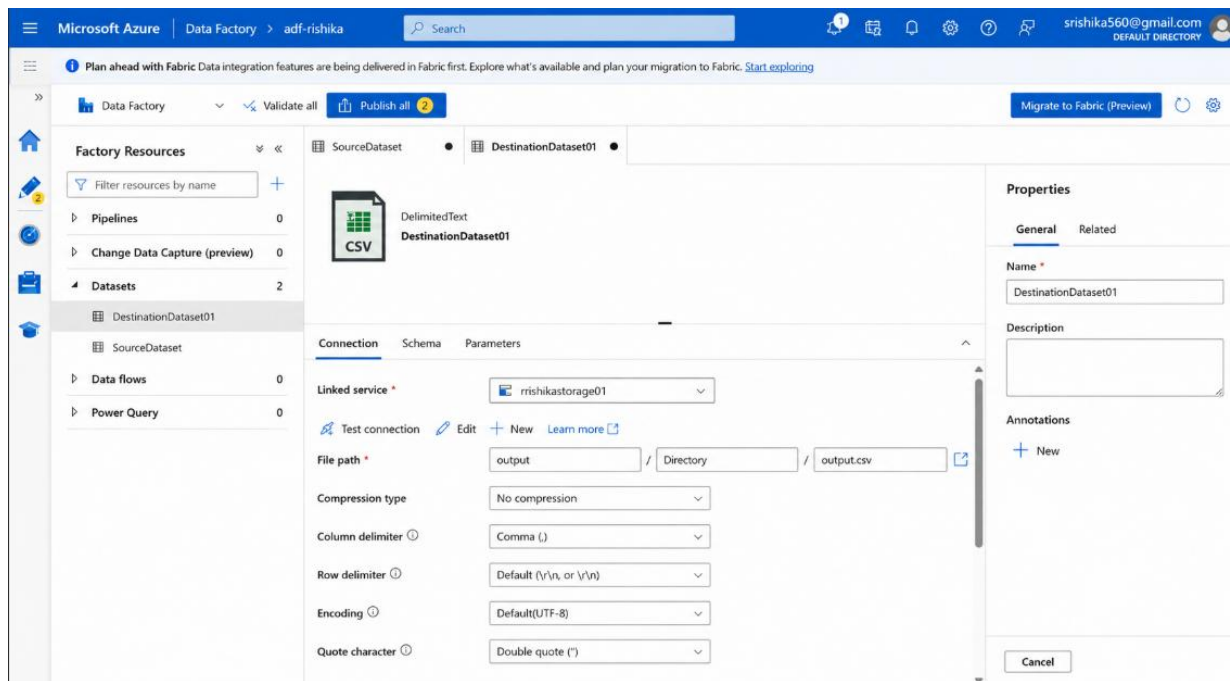
This dataset represents the source file that will be processed by the pipeline.



Output Dataset

- **Name:** DestinationDataset
- **Container:** output
- **File:** output.csv

This dataset specifies the location where the copied data will be stored.



Implementing the Get Metadata Activity

The Get Metadata activity is used to obtain information about a file or dataset before performing additional operations.

In this assignment, the activity was configured with **SourceDataset**, and the **Exists** property was selected to confirm the availability of the source file before starting the copy operation.

Benefits of Using Get Metadata

- Verifies whether the source file exists.
- Confirms data availability before processing.
- Improves pipeline reliability by preventing execution errors.

Step 4 – Building the Data Pipeline

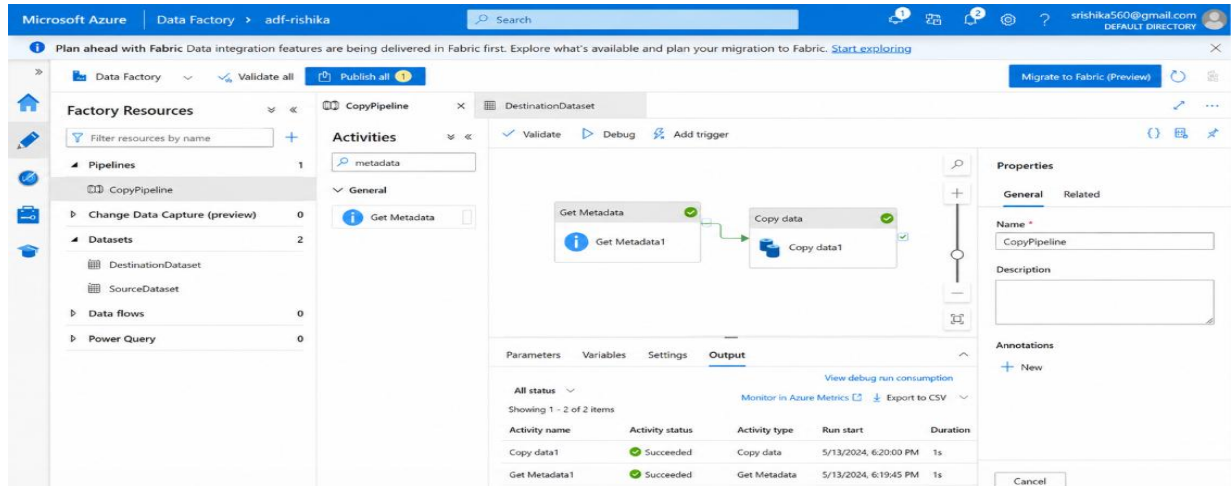
Objective

The purpose of this task is to create an Azure Data Factory pipeline that transfers data from the source dataset to the destination dataset using the Copy Data activity.

Step 4 – Building the Data Pipeline

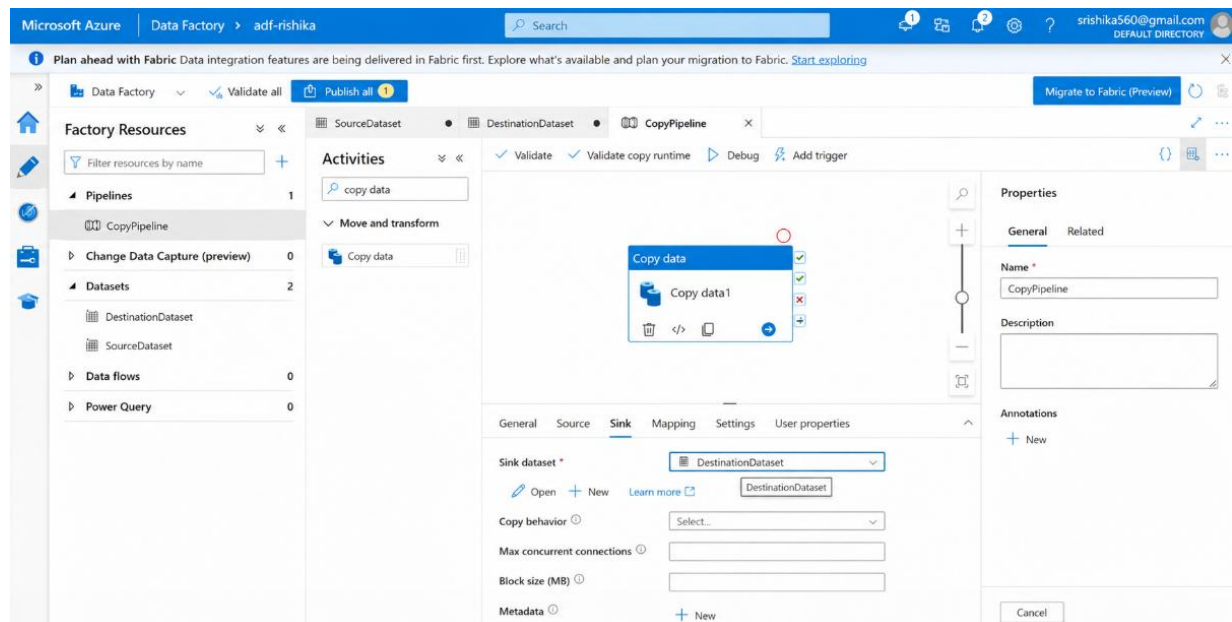
Objective

The purpose of this task is to create an Azure Data Factory pipeline that transfers data from the source dataset to the destination dataset using the Copy Data activity.



Pipeline Configuration

A pipeline named **CopyPipeline** was developed in Azure Data Factory. The Copy Data activity was configured to read data from **SourceDataset** in the input container and copy it to **DestinationDataset** in the output container.



Step 5 – Running the Pipeline

Objective

The objective of this task is to execute the Azure Data Factory pipeline and ensure that the data transfer process is completed successfully.

Pipeline Execution Process

The pipeline was executed using the **Debug** option in Azure Data Factory. During execution, the data was copied from the **SourceDataset** in the input container to the **DestinationDataset** in the output container.

The screenshot displays the Azure Data Factory interface for a pipeline named 'CopyPipeline'. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, showing 'CopyPipeline' with a count of 1. Below it, 'Datasets' are listed: 'DestinationDataset' and 'SourceDataset'. The main canvas shows the pipeline design with a 'Copy data' activity connected to 'Copy data1'. The 'Activities' pane on the left lists 'copy data' and 'Copy data1'. The 'Properties' pane on the right shows the 'General' tab with the name 'CopyPipeline' and a description field. The 'Output' pane at the bottom shows the pipeline status as 'Succeeded' with a green checkmark. Below this, a table lists the activity status:

Activity name	Activity status	Activity tc...	Run start	Du
Copy data1	Succeeded	Copy data	6/14/2026, 6:26:36 PM	19s

Step 6 – Configuring IAM Permissions

Objective

The objective of this task is to manage access permissions by assigning appropriate IAM (Identity and Access Management) roles to users and Azure Data Factory.

Role Configuration

The **Reader** role was assigned to provide read-only access to the storage resources. Additionally, the **Storage Blob Data Contributor** role was granted to Azure Data Factory, enabling it to read and write data in Azure Blob Storage and ensuring successful pipeline execution.

Microsoft Azure

Data Factory > adf-rishika

Search

1

2

srishika560@gmail.com
DEFAULT DIRECTORY

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Data Factory

Validate all

Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines1

CopyPipeline

Change Data Capture (preview)0

Datasets2

DestinationDataset

SourceDataset

Data flows0

Power Query0

CopyPipeline

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metadata

General

Get Metadata

DestinationDataset

Validate

Debug

Add trigger

Get Metadata

Copy data

Parameters

Variables

Settings

Output

All status

Showing 1 - 2 of 2 items

Activity name	Activity status	Activity type	Run start	Duration
Copy data1	Succeeded	Copy data	5/13/2024, 6:20:00 PM	1s
Get Metadata1	Succeeded	Get Metadata	5/13/2024, 6:19:45 PM	1s

View debug run consumption

Monitor in Azure Metrics

Export to CSV

Properties

General

Related

Name

CopyPipeline

Description

Annotations

New

Cancel